

Daniel Rodríguez Carrión

R&D, Embedded Systems and IoT Engineer

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Professional Profile

Electronics, Robotics and Mechatronics Engineer with a Master's degree in Mechatronics Engineering. Experience in R&D, cybersecurity applied to embedded devices, firmware, sensing systems and IoT infrastructures, working with C/C++, Python, ESP32, FreeRTOS, Linux and test automation. A product-oriented profile with experience in prototyping, technical validation and multidisciplinary collaboration in international environments.

Technologies, level and experience

Area	Stack / technologies	Level	Approx. years
Firmware & Embedded	C, C++, ESP32, ESP8266, FreeRTOS, OTA, RISC-V	Advanced	4 years
Software & scripting	Python, Bash, CMake, Makefiles	Advanced	4 years
IoT & communications	MQTT, Node-RED, BLE, Ethernet, I2C, JTAG	Advanced	3 years
Systems & infrastructure	Linux/Ubuntu, VMs, Raspberry Pi, Docker	Intermediate-Advanced	4 years
Embedded security	Secure boot, secure storage, cryptography, IoT evaluation	Advanced	2 years
CAD and 3D design	CATIA V5, 3D modeling, mechatronic prototyping	Intermediate	2 years
Data and tooling	data analysis, charting, technical automation	Intermediate	3 years

Types of projects I am looking for

- Technology products: firmware, embedded devices, IoT, sensing systems, robotics and edge computing.
- Engineering and R&D services: prototyping, validation, test automation and hardware-software integration.
- Projects with a cloud/data component linked to IoT: data acquisition, monitoring, data pipelines and technical tooling.

Work model and availability

Preferred work model	Remote or hybrid
Location	Spain
Availability	Available to join within 2-4 weeks
International environment	Yes, comfortable working in technical English

Professional Experience

Cybersecurity and R&D Engineer

DEKRA

01/2025 – 02/2026

- Member of the evaluation team for Google pKVM Android SESIP Level 5 certification.
- Security evaluation of IoT devices and interfaces such as BLE, LTE, Ethernet/MACsec, I2C and JTAG.
- Validation of critical functions such as secure boot, secure firmware upgrade, secure debug and secure storage.
- Development and execution of test campaigns in Linux/VM environments with Python and C/C++ automation.

Senior Mechatronic & Aerial Robotics Engineer

AVERON

01/2025 – 05/2025

- Technical collaboration in the development of aerial robotics solutions and autonomous drones.

- Conceptual and mechanical drone design, 3D modeling and mechatronic integration.

Technical Staff and Researcher

University of Málaga

02/2022 – 09/2024

- Technical development in the HortSost II project: sensing systems, instrumentation, ESP32 firmware and data management with MQTT and Node-RED.
- Research on self-configurable sensor infrastructures for edge computing.
- Publication and conference presentation on a wireless sensor network for edge computing.

Technical Staff

SITER 2000

06/2019 – 01/2020

- Support in consulting and deployment studies for GSM/DCS/UMTS mobile networks and FTTH.
- Review of engineering documentation and as-built materials for access point installation.

Education

- Master's Degree in Mechatronics Engineering — University of Málaga (2023)
- Bachelor's Degree in Electronics, Robotics and Mechatronics Engineering — University of Seville / Málaga (2021)
- Project Management for Photovoltaic Installations — Euroinnova (2023)

Certifications and Languages

- CATIA V5 - Part Design (Associate), Dassault Systèmes (2024)
- RPAS Pilot and Radio Operator, Aerocámaras (2019)
- Native Spanish, English B2 (FCE Cambridge)

Publication

Self-configurable wireless sensor network for edge computing, Spanish Congress of Informatics (CEDI), A Coruña, 2024.